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Selective laser melting-wire arc additive manufacturing hybrid fabrication of Ti-6Al-4V alloy: Microstructure and mechanical properties

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ABSTRACT

Selective laser melting (SLM) and wire arc additive manufacturing (WAAM) are two representative metal additive manufacturing technologies that fabricate the net shaping of small complex parts and near-net shaping of large components. To explore the feasibility of SLM-WAAM hybrid fabricating large and complex components, the preliminary hybrid SLM-WAAM process of Ti-6Al-4V is investigated. In this study, horizontal (H-) and vertical (V-) SLM samples are used as substrate for WAAM process to analyze microstructure and tensile properties, and to be compared with that of SLM samples and WAAM samples. The results show that Ti-6Al-4V sample formed by hybrid fabrication consists of three typical zones: SLM zone, WAAM zone and interface zone. The fine and short β columnar grains consisting of primarily martensite α´ exist in SLM zone, then these fine grains grow coarse in the about 3 mm-thick interface zone because of the repeated heating process, in contrast, the coarse β columnar grains awash with α lamellae lead to epitaxial growth in WAAM zone. There is a good metallurgical bonding in the interface zone of both H-SLM-WAAM and V-SLM-WAAM specimens and all the fractures are located in the WAAM zone during the tensile test. The mean yield strength and ultimate tensile strength of the H-SLM-WAAM specimens are 850 MPa and 905 MPa, respectively. In comparison, the V-SLM-WAAM samples exhibited slightly higher tensile strengths; the mean yield strength and ultimate tensile strength are 890 MPa and 995 MPa, respectively. The elongations of different directions are both more than 10%, satisfying the requirements of engineering applications. These tensile properties of SLM-WAAM samples are comparable with or even better than that of WAAM samples.

1. Introduction

Additive manufacturing (AM) is a process of stacking materials layer upon layer to fabricate components from 3D model data [1–3]. Within the last decades several technologies for additive manufacturing of metal parts have been developed rapidly due to its high material utilization and high geometrical freedom, which is particular suitable for the production of structural parts in aviation, aerospace, and medical industries [4,5]. These metallic additive manufacturing technologies can be divided into two types (although there are many more): powder bed-based additive manufacturing [6,7] and powder/wire-feed additive manufacturing [8–10].

The powder bed-based additive manufacturing techniques, such as selective laser melting (SLM) [11,12] and electron beam melting (EBM) [13,14], fabricate functional complex parts by selectively melting successive layers of fine powder using a laser or electron beam. These techniques can generate metal parts with fine surface roughness, high density, high mechanical properties, and even arbitrary complex structures. However, they are limited by the high cost, low build rate and small scale.

Compared with powder bed-based additive manufacturing technologies, powder/wire-feed additive manufacturing techniques, such as wire arc additive manufacturing (WAAM) [15], laser metal wire deposition (LMWD) [16] and laser melting deposition (LMD) [17] produce near-net shaped components by utilizing arc or laser welding, which are similar to a conventional welding process. These techniques are well suited for manufacturing parts of high deposition rate and large volumes; however, the fabricated product is limited by the geometrical freedom due to the lack of support of loose powder during manufacturing and the inadequate geometrical accuracy and the surface roughness, so a subsequent machining of the built part is required.

Generally, there are distinct advantages to each type of additive manufacturing system. The powder bed-based additive manufacturing is generally suitable for manufacturing small, complex parts, while the wire-feed additive manufacturing has superior adaptability of fabricating large, simple structure parts. When manufacturing large-size

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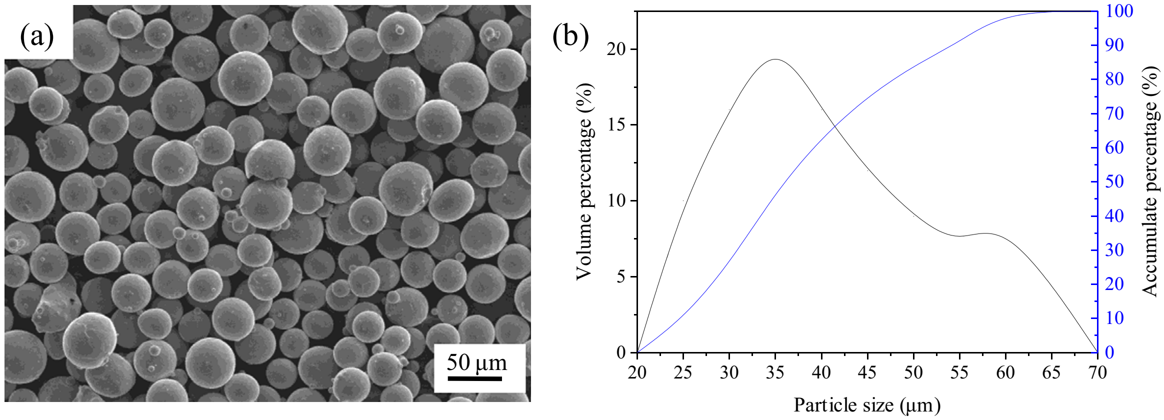

Fig. 1. (a) Ti-6Al-4V powder morphology; (b) Particle size distribution.

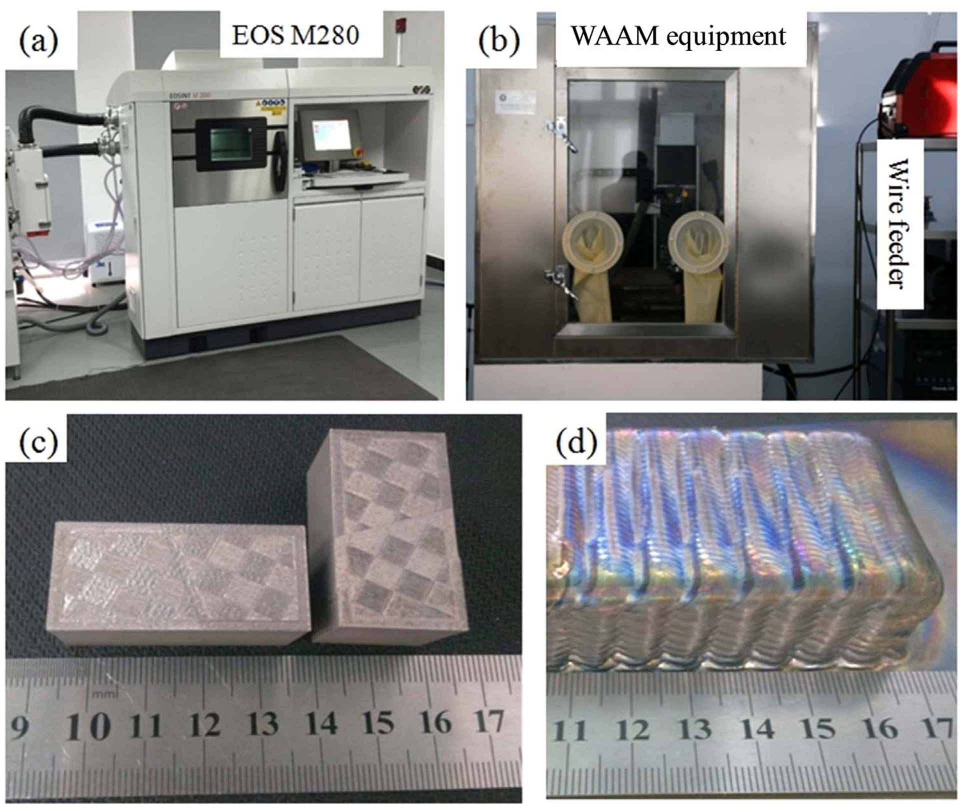

Fig. 2. (a) SLM experimental equipment; (b) WAAM experimental equipment; (c) As-built samples produced in SLM; (d) As-built samples produced in WAAM.

Table 1
Main experimental parameters of selective laser melting.

Laser power	310 W
Laser spot diameter	150 µm
Scanning speed	1100 mm/s
Layer thickness	60 µm
Hatch space	180 µm
Scanning strategy	meander

Table 2
Deposition parameters of wire arc additive manufacturing.

Wire feed speed	100 cm/min
Welding speed	120 mm/min
Peak current	180 A
Peak time ratio	30%
Base current ratio	30%
Pulse frequency	1.8 Hz
Gas flow rate	20 L/min
Arc length	3 mm

complicated components, the advantages of the two techniques can be well combined by adding special features of powder/wire-feed additive manufacturing to the base fabricated by the powder bed-based additive manufacturing technique. This is a more flexible fabrication process based on additive manufacturing techniques, named the hybrid fabrication (HF) technique [18]. Wang et al. explored the microstructure and mechanical properties of hybrid fabricated 1Cr12Ni2WMoVNb steel by laser melting deposition (LMD) on a wrought bar of same material. Results showed that the hybrid manufactured sample consisted of three typical zones: the laser deposited zone, the wrought substrate zone and the bonding zone. The sample formed by hybrid fabrication had a good metallurgical bonding between the laser deposited zone and the wrought substrate, and fracture occurred in the wrought substrate zone during the tensile

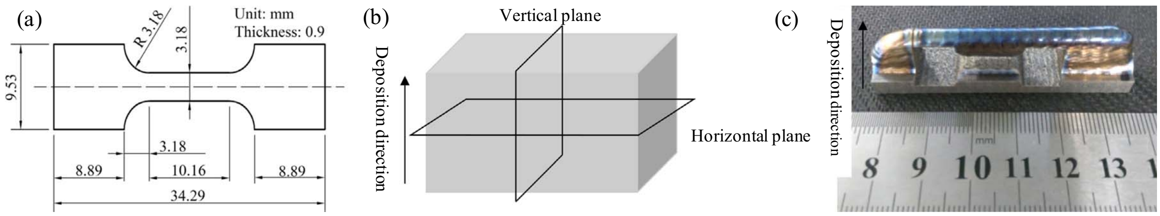

Fig. 3. Parameters of tensile samples, (a) the size of tensile samples, (b) the direction of tensile samples of SLM, (c) the macroscopic images and the direction of tensile samples of WAAM.

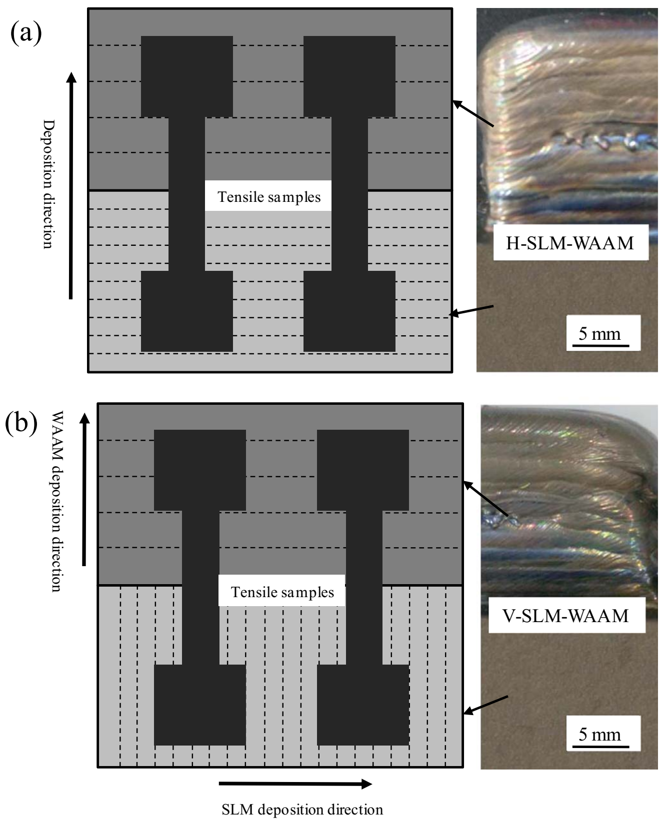

Fig. 4. Schematic diagram of hybrid process strategies and corresponding images: (a) horizontal SLM-WAAM, (b) vertical SLM-WAAM.

test [19]. In addition, Zhu et al. reported the hybrid fabricated TC11 sample had good mechanical properties that fell between the higher strength LMD sample and the lower strength wrought sample, and the fracture of hybrid sample occurred in the substrate in tensile testing, meaning that the bonding zone had better mechanical properties than the substrate [20]. There is no doubt that the HF technique can further increase the utilization ratio of materials and reduce the building time and costs.

At present, aerospace structural titanium alloy components are mostly fabricated by conventional costly casting and forged processes at a low material utilization ratio. Especially for Ti-6Al-4V alloy, it is widely used in the aerospace industry because of its high strength-to-weight ratio [21]. The HF technique provides a new direction for producing large and complex Ti-6Al-4V components. Liu et al. proposed LMD-SLM hybrid manufacturing processes in order to achieve subscale complex structures. Samples with high density and good mechanical strength could be fabricated, and tensile strength and elongation of the hybrid fabricated Ti-6Al-4V samples could reach respectively 918 MPa and 11% respectively [22]. In comparison to LMD technique, the WAAM process has some remarkable advantages: lower cost and higher efficient. Therefore, the SLM-WAAM hybrid fabrication technique can further reduce the building time and costs. Welding is another cost-reduction manufacturing technology, this technique can also be applied for welding titanium structures and high mechanical properties have been successfully obtained [23,24]. However, it is difficult to apply to the conditions of irregular welding shapes or high thickness joints. Available welding machine is only capable of welding full penetration joint below 20 mm in titanium alloy [25,26]. So the integration of SLM and WAAM components in a welding assembly is limited in these features. In contrast, the HF technique is no longer restricted by the shape and interface thickness. During the SLM-WAAM hybrid process, at first, the large component needs to be well designed and divided into two parts, the complicated part and the simple part; then, the complicated part is finished by SLM and retained as a substrate of the WAAM process; finally, the rest part is completed by WAAM.

One common concern of the hybrid fabrication technique is the mechanical property of the finished part, which is dominantly dependent on the microstructure of the final part especially in the interface zone. In this study, the preliminary feasibility evaluation of hybrid SLM-WAAM fabricating Ti-6Al-4V is investigated. In order to obtain a deep understanding of the SLM-WAAM hybrid fabricating process of Ti-6Al-4V alloy, the microstructure and tensile properties at room temperature of the SLM samples, WAAM samples and hybrid fabricated samples are investigated. And the microstructure evolution in the interface zone during the hybrid fabricating process is also discussed.

2. Experimental procedures

Ti-6Al-4V powder supplied by AVIC BIAM was used for SLM. Fig. 1a shows the powder morphology under a scanning electron microscope (SEM). Fig. 1b shows the distribution with an average particle size of 36 µm (d10:25 µm, d90:55 µm). For WAAM, Ti-6Al-4V wire of 1.2 mm in diameter was used.

SLM and WAAM experiments were respectively conducted using an EOS M280 SLM machine (Fig. 2a) and a wire arc additive manufacturing equipment which was developed by Beijing Institute of Technology (Fig. 2b). The as-built Ti-6Al-4V samples (Fig. 2c and d) were using the optimal combinations of process parameters, as shown in Table 1 (for SLM) and Table 2 (for WAAM). In the SLM-WAAM hybrid processed experiment, the SLM samples with the size of 30×30×30 mm were firstly built, then these specimens were used as the substrate for WAAM experiment after milling the top surfaces and about 30×30×20 mm blocks were fabricated by WAAM. In consideration of the practical application of the SLM-WAAM hybrid manufacturing, the WAAM parts could be added at the top/side surfaces of the SLM part in which the SLM part is used as substrate. So these two strategies of H-SLM-WAAM and V-SLM-WAAM need to be taken into account, as shown in Fig. 4.

No heat treatment was performed on any of the samples produced in this study. For the tensile properties test, the shape and dimensions of tensile specimens are exhibited in Fig. 3a. The SLM tensile samples were extracted in two directions: parallel and perpendicular to the building direction (Fig. 3b) and the WAAM tensile sample was extracted parallel to the deposition direction using wire electrical discharge machining (Fig. 3c). The SLM-WAAM tensile samples were extracted according to Fig. 4. At least three tensile samples were

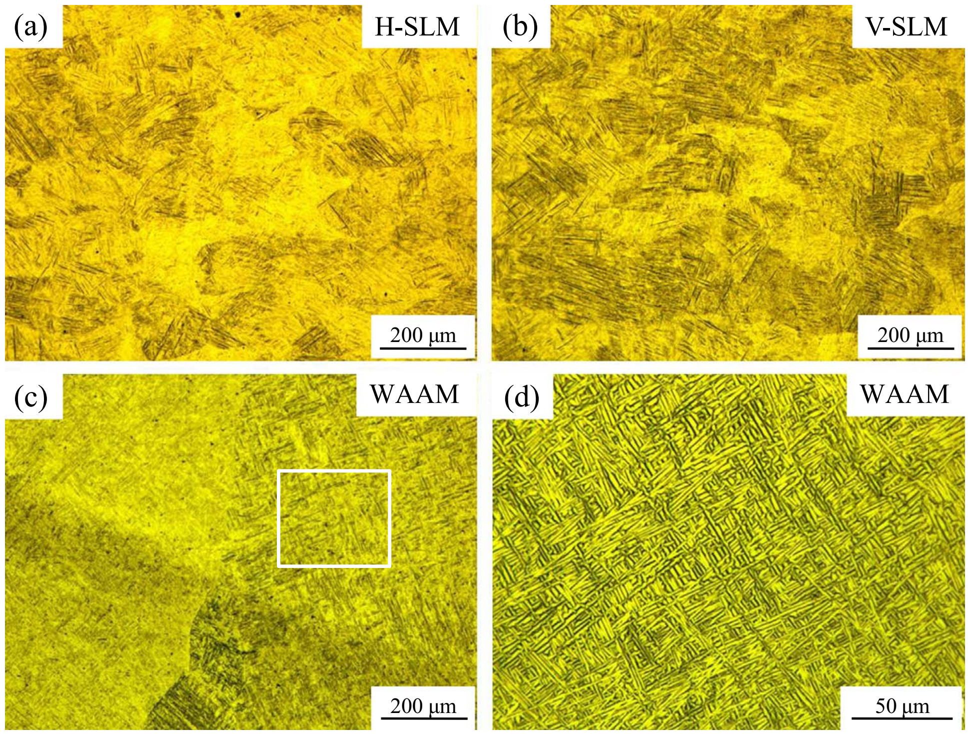

Fig. 5. (a) Horizontal section microstructure of SLM sample, (b) Vertical section microstructure of SLM sample, (c) Vertical section microstructure of WAAM sample, (d) Enlarged view from the boxed region in "c".

prepared for each condition. The tensile tests were carried out with an Instron testing machine at room temperature. The displacement rate of cross head was 0.01 mm/s, and a dynamic strain gauge extensometer was applied to record the strain. For the metallographic analysis, metallographic specimens were prepared using standard grinding and polishing procedures, and etched by Kroll's reagent (90 ml H_2O, 6 ml HNO_3, and 2 ml HF). A Leica DM4000M optical microscope was used for microstructural examination. The fracture morphology of fractured specimens was analyzed by a JEOL-JSM6490 scanning electron microscope (SEM). The element content was recorded using electron microprobe analysis (EMPA) measurement (JEOL JXA 8100).

3. Results and discussion

3.1. Microstructure

As we know, the microstructure of Ti-6Al-4V typically consists of a mixture of α and β phases at room temperature. However, representative optical micrographs of the SLM and WAAM samples show a completely different microstructure, as shown in Fig. 5. For SLM samples, the microstructures of horizontal section and vertical section are both short β columnar grains with the width about 150 μm and height about 200 μm, which consist of primarily acicular martensite (α´), as can be seen in Fig. 5a and b. In contrast, the coarse β columnar grains can be clearly observed in the WAAM samples, and the width of the columnar grain can reach 500–700 μm as shown in Fig. 5c. The α lamellae are arranged in a basket weave structure with different sizes and orientations and formed within the columnar grains as can be seen in Fig. 5d. These microstructures are typical of SLM and WAAM samples in Ti-6Al-4V and have been discussed by many earlier investigations [9,27–30].

Many studies indicate that microstructure evolution of Ti-6Al-4V primarily depends on the cooling rate when the β transforms to α as it cools through the transus temperature [8,21,31,32]. The martensitic α´ transformation of Ti-6Al-4V can be formed under high cooling rates (more than 410 ℃/s) [33], and the lamellar α microstructure can be formed at the intermediate cooling rates (between 20 ℃/s and 410 ℃/s) [8]. It is clear that the SLM process with a laser beam diameter of 100–150 μm focused on the powder bed to generate a tiny molten pool with the diameter of about 200 μm, and yet the WAAM process with wire diameter of 1.2 mm produce a larger molten pool with a size of about 4 mm, which will inevitably lead to an increase of the thermo-capillary convection whereas result in a decrease of the cooling rate. Therefore, the SLM process has higher cooling rates above the critical cooling rate of 410 ℃/s, which cause the transformation of α to α´ and the β grain refinement.

Fig. 6 shows the cross-sectional microstructure of the interfaces in the SLM-WAAM hybrid-processed samples. The hybrid fabricated Ti-6Al-4V sample consists of three typical zones: SLM zone, WAAM zone, and interface zone. There are no defects in the bonding interfaces of both the H-SLM-WAAM and V-SLM-WAAM specimens, which indicate that the hybrid process has a dense interface microstructure, and obviously the interface region is not a defective cluster zone. The β grain and α-lath widths were examined by analyzing micrographs and quantified using ImageJ software. As shown in Fig. 6a and b, the short β columnar grains with the width of 150 μm in SLM zone grow coarse to a size of 250 μm in the interface zone, then the columnar β grains preferentially grow along the deposition direction and the width increases to 700 μm in the WAAM zone. In both the H-SLM-WAAM and V-SLM-WAAM samples, statistically significant differences can also be found in the mean α-lath widths of different regions. In SLM zone, the average of acicular α width is 0.68 μm (see Fig. 6c); the α-lath width is slightly larger, averaging near 0.85 μm in the 3 mm-thick interface zone (see Fig. 6d and e), which was caused by heat effect while the Ti-6Al-4V wire was deposited on the SLM-substrate during the WAAM process; in the WAAM zone, the lath width averaged 1.32 μm (see Fig. 6f). Fig. 7 illustrates the microstructure evolution processes during the hybrid process. The SLM samples act as the substrate of WAAM and the substrate surface is remelted by arc during the deposition of first layer. Thus the fine β columnar grains grow coarse

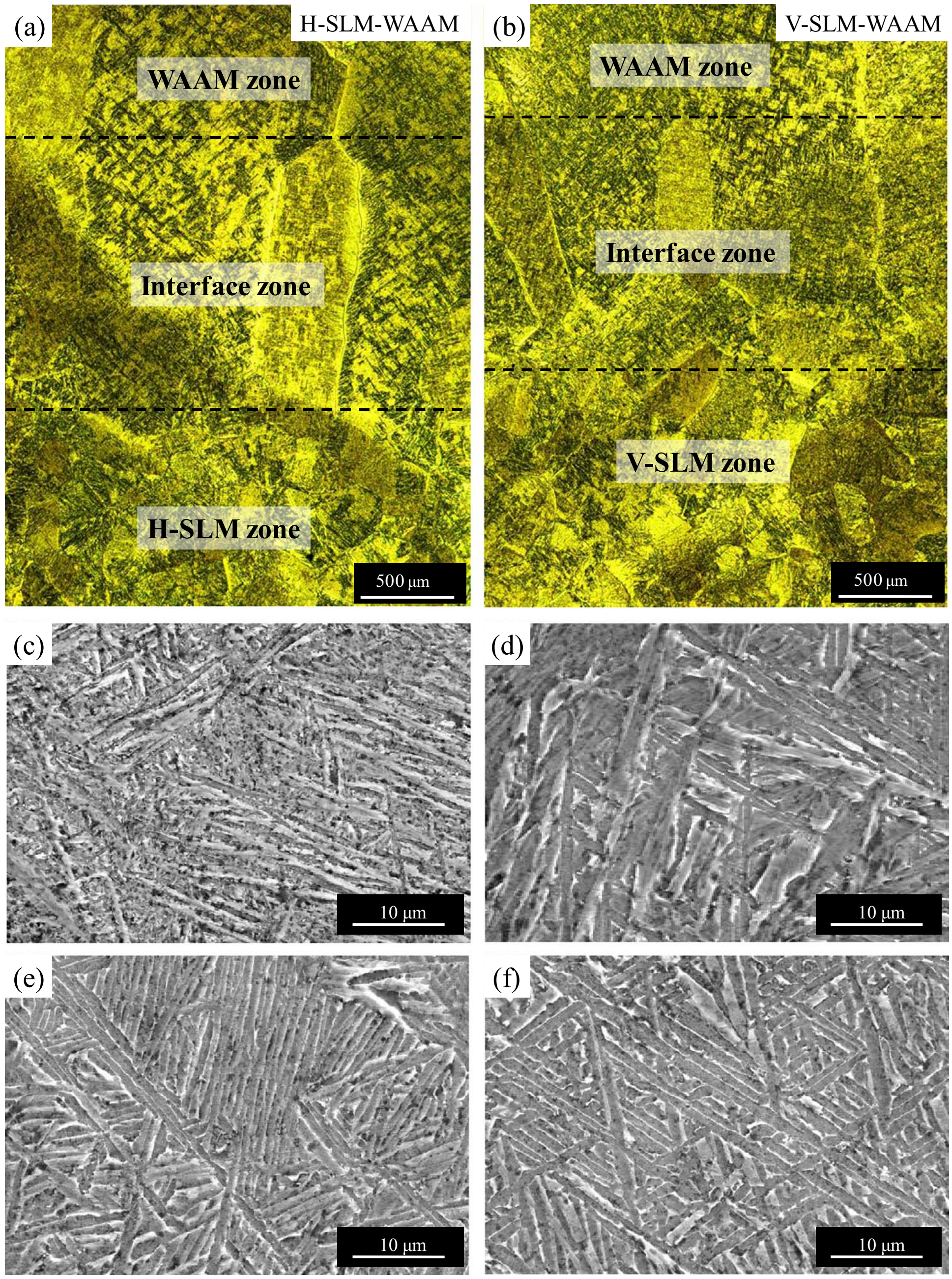

Fig. 6. Microstructures of hybrid-processed samples: (a) horizontal SLM-WAAM, (b) vertical SLM-WAAM, (c) SLM zone, (*d*-e) Interface zone, (f) WAAM zone.

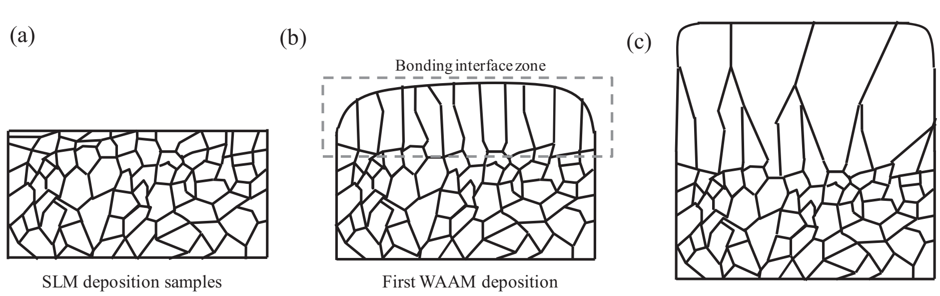

Fig. 7. Schematic illustration for the formation process of β grain morphology during SLM-WAAM hybrid forming process.

in the interface zone because of the repeated heating process (Fig. 7b). Then, with the depositing of many layers, the columnar β grains will lead to epitaxial growth and preferred growth from the interface zone to the deposition area due to the high temperature gradient that exists along the building direction in the WAAM process, as schematically shown in Fig. 7c.

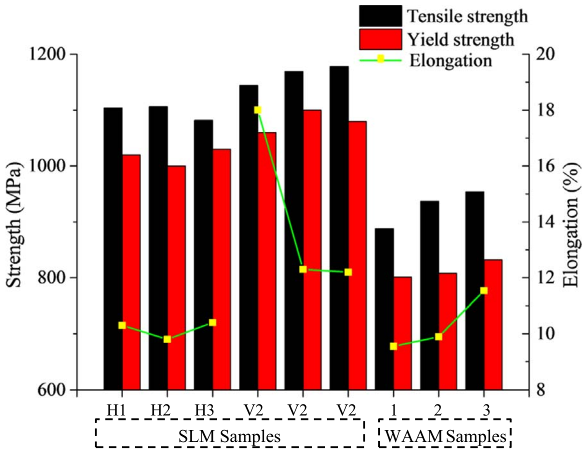

Fig. 8. Tensile properties of SLM samples and WAAM samples; tensile strength and yield strength and elongation.

3.2. Tensile properties

Tensile test results of H-SLM, V-SLM and WAAM-produced Ti-6Al-4V samples are summarized in Fig. 8. Each group included three duplicate tests. The yield strength, ultimate tensile strength and elongation were obtained as direct output from the tensile testing machine. The strain measurements were measured using a video extensometer by capturing continuous images of the specimen during the test. A corresponding difference in mechanical properties between SLM-processed and WAAM-processed samples can be seen. The mean yield strength, ultimate tensile strength and elongation of the horizontal SLM specimens are 1020 MPa, 1098 MPa and 10.2%, respectively. In comparison, the samples tested from the vertical direction exhibited slightly higher tensile strengths, with the mean yield strength of 1050 MPa and ultimate tensile strength of 1163 MPa. The tensile strength values are only moderately anisotropic. However, greater asymmetry can be seen in the ductility value, the mean elongation of the vertical SLM specimens is 14.1% which is much better than that measured in the horizontal direction. This is reasonable that the columnar β grains grow parallel to the building direction; it is loaded transverse to the prior-β grain boundaries when measured in the horizontal direction, apparently this would be expected to promote premature failure through the grain boundary [30]. By contrast, the WAAM Ti-6Al-4V samples have the lower strength and ductility, the mean yield strength, ultimate tensile strength and elongation are 814 MPa, 927 MPa and 10.5%, respectively. Overall, the tensile properties of the WAAM-produced Ti-6Al-4V samples are slightly lower when compared to those of the SLM samples.

These results are primarily attributed to the differences in the microstructure (see in Fig. 5). The mechanical property of Ti-6Al-4V is directly affected by the average grain size according to the Hall-Petch equation [31,34]. With the size of β grains decreasing, the mechanical properties including yield strength, ultimate tensile strength and elongation will be improved. Therefore, the short β columnar grains exhibited in the SLM samples (Fig. 5a and b) result in high tensile strengths and ductility; on the contrary, the coarse β columnar grains in the WAAM samples (Fig. 5c) lead to the reduction of tensile properties. However, majorities of acicular martensites α′ are formed in SLM-produced Ti-6Al-4V samples, which lead to high mechanical strength but low ductility [33,35]. Hence the tensile strengths further increase and the elongation slightly decreases in SLM samples compared to the lamellar α microstructure existed in WAAM-produced Ti-6Al-4V samples (Fig. 5d). On the whole, the yield strength and ultimate tensile strength values of SLM-produced Ti-6Al-4V samples are much higher than the values of WAAM-produced Ti-6Al-4V samples; the elongation values of H-SLM and WAAM specimens are similar and both lower than the value of V-SLM specimen. The tensile properties of Fig. 8 are consistent with the observed microstructure as shown in Fig. 5.

Representative fractographs of the tensile fracture surface of H-SLM, V-SLM and WAAM-produced Ti-6Al-4V test specimens are shown in Fig. 9. The fracture surfaces of all the SLM and WAAM test specimens are characterized by dimple-like structures, which are indicative of ductile rupture. The difference of fracture surfaces is the feature of dimples, which can reflect their mechanical properties. Many researches indicated that the small size of dimples lead to high tensile strength [8] and low tensile ductility [36], and the shallow dimples reflect the drop in elongation for the Ti-6Al-4V samples [37]. Small dimples with the size of 1–3 μm can be observed on both H-SLM and V-SLM fracture surfaces (Fig. 9a and b), but the dimples in H-SLM samples are shallower than those in V-SLM samples. Consequently, the elongation of H-SLM samples is lower than that of the V-SLM samples, and the tensile strength are basically flat. The fracture surfaces of WAAM samples (Fig. 9c) also exhibit similar morphology to those of V-SLM samples, which are bestrewed with dimples. Fig. 9b shows small and shallow dimples on V-SLM fracture surface in contrast to large and deep dimples for the WAAM product shown in Fig. 9c. The dimple size of the WAAM samples ranges from 2 to 6 μm, or approximately 50% larger than that of the V-SLM samples. Therefore, the tensile strength and yield strength of the WAAM samples are inferior to those of the V-SLM samples, and the elongation is also lower than that of V-SLM samples. It is likely that the combination of micropores and residual stress result in the low ductility [35,38] or the data dispersion and sample quantity limitation. An in-depth study needs to be carried out in a subsequent experiment. On the whole, the fractographs are consistent with tensile properties shown in Fig. 8.

Fig. 10 shows hybrid-processed plate tensile specimens. All the fracture surfaces have significant neck contractions, as shown in Fig. 10a and b. After the polishing and etching the fracture plate tensile specimens, element segregation in the interface between two adjacent layers results in a different corrosion effect (see Fig. 10c and d). It can be seen all the fractures are located in the WAAM zone. This indicates that the interface zone has a good metallurgical bonding effect and good mechanical properties, which performed better than WAAM zone for strength, ductility and fatigue crack initiation. The morphol-

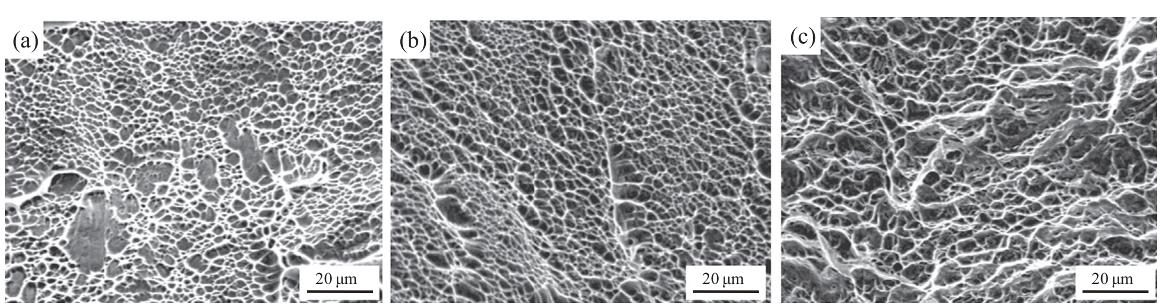

Fig. 9. SEM micrographs of tensile fracture surfaces of (a) Horizontal SLM-produced tensile sample, (b) Vertical SLM-produced tensile sample, (c) Vertical WAAM-produced tensile sample.

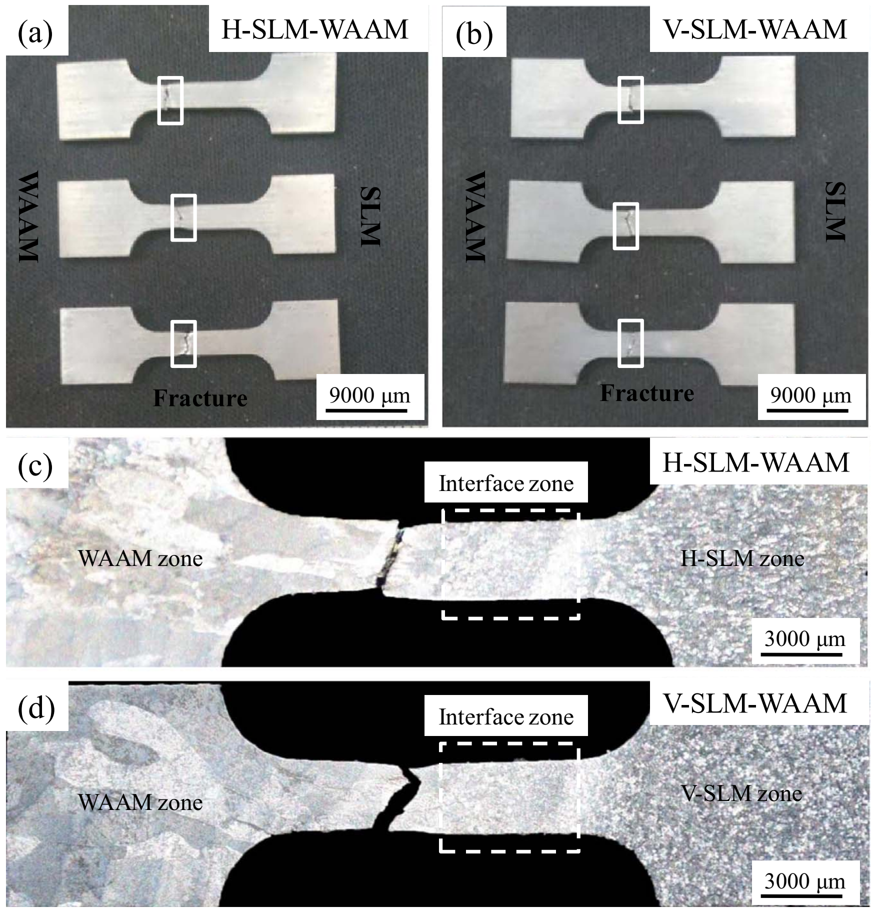

Fig. 10. Hybrid-processed plate tensile samples: (a) horizontal SLM-WAAM, (b) vertical SLM-WAAM, (c) Optical macrostructure image of the fracture plate tensile specimens.

Table 3
Chemical compositions of original materials and SLM-WAAM samples.

Material	Sample	Form/region	Al (wt%)	Ti (wt%)	V (wt%)
Ti–6Al-4V	materials	Powder	6.25	89.51	4.24
		Wire	6.19	89.90	3.91
	SLM-WAAM	SLM zone	5.83	90.07	3.91
		Interface zone	5.65	90.13	4.22
		WAAM zone	5.74	90.29	3.97

ogy of the grains is gradually changed from uniformly distributed granular grains in the SLM zone and coarsened grains in the interface zone to epitaxially coarse columnar grains in the WAAM area. Because of the high-temperature gradient caused by the arc deposition process, the columnar grains grow along the deposition direction in SLM substrate, and about 3 mm-thick grain deformation-increased interface zone can be observed.

The values of Ti, Al, V elements of original materials and final products were measured by EMPA measurement and each region measured five points then averaged the values, the results were

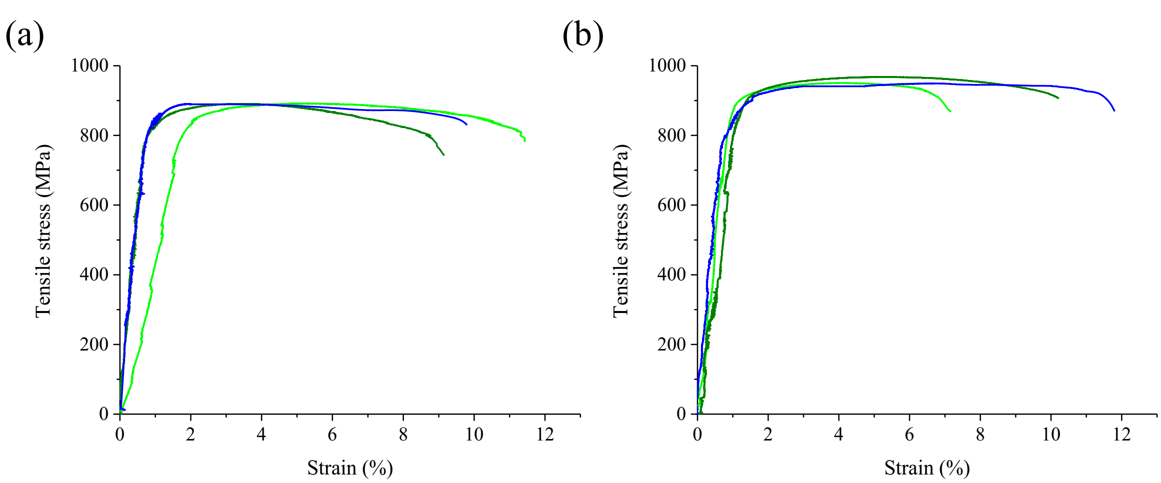

Fig. 11. Stress-strain plots of SLM-WAAM hybrid-processed samples (a) horizontal SLM-WAAM, (b) vertical SLM-WAAM.

Table 4
Tensile properties of Ti-6Al-4V fabricated by different processes.

Process	Yield stress (Offset 0.2%), MPa	Ultimate tensile stress, MPa	Elongation, %	References
SLM	1110	1267	7.28	[33]
SLM	1098	1237	8.8	[40]
WAAM	950	1033	11.7	[30]
WAAM	909	988	7.5	[41]
LBW	942	974	10.03	[32]
EBW	922	967	9.7	[25]
GTAW	869	960	5.6	[26]
H-SLM-WAAM	850(SD=1)	890(SD=1)	10.2(SD=0.8)	In this study
V-SLM-WAAM	905(SD=5)	960(SD=5)	10.0(SD=1.7)	

SD: standard deviation

processed by normalization, as shown in Table 3. The amounts of Ti, Al and V are constant at different regions of the final products, and a slight variance between the original materials and final products (within +0.60/−0.78 wt%) can be found, which is because the overheating resulting in the evaporation of Al element from the Ti-6Al-4V during the high energy input process [39]. However, the changes are fairly minor, and these values can meet the requirements for this material, specified by ASTM B348-13 (Grade 5 Ti-6Al-4V).

As shown in Fig. 11, ultimate tensile strength of the three H-SLM-WAAM tensile specimens ranges from 889 MPa to 891 MPa, yield strength ranges from 849 MPa to 850 MPa, and elongation ranges from 9.2–11.4%, while the ultimate tensile strength of the three tensile V-SLM-WAAM specimens ranges from 950 MPa to 960 MPa, yield strength ranges from 900 MPa to 910 MPa, and elongation ranges from 8.1–11.9%. The values of the tensile properties of H-SLM-WAAM and V-SLM-WAAM samples are relatively stable. The average values of the yield strength are 850 MPa and 905 MPa, the average values of the ultimate tensile strength are 890 MPa and 955 MPa, while the average values of the elongation are 10.2% and 10.1%, respectively. It can be seen that the hybrid-forming samples exhibit higher tensile strengths of the V-SLM-WAAM sample than that of the H-SLM-WAAM; this is because the SLM samples have higher tensile strengths in the vertical direction, which affects the overall tensile properties and improves the tensile intensities. The influence of SLM H/V direction on the elongation of hybrid-forming samples is slight, and the elongation of different directions is more than 10%, satisfying the requirements of engineering applications. This is because all the fractures are located in the WAAM area, thus the elongation of hybrid-forming samples mainly reflects the performance of the WAAM area with a relatively low plastic property of 10.5%. Overall, the results indicate that the tensile properties of the hybrid-processed samples with H/V direction are isotropic. And the tensile properties of SLM-WAAM samples are comparable with or even better than the WAAM samples. Table 4 summarizes the tensile properties of Ti-6Al-4V fabricated by different processes. The mechanical properties of SLM-WAAM fabricated samples are compared to that of Ti-6Al-4V samples formed by SLM [33,40], WAAM [30,41], laser beam welding (LBW) [32], electron beam welding (EBW) [25] and gas tungsten arc welding (GTAW) [26]. The results indicate that good mechanical properties can be obtained of the SLM-WAAM samples, and these values are comparable to the data that given in the literatures, but the SLM samples present higher tensile strengths because of the existence of martensite.

Shear lip can be observed on both H-SLM-WAAM and V-SLM-WAAM samples as shown in Fig. 12a and b, and the enlarged fracture surfaces show ductile dimpled rupture features (Fig. 12c and d). The fracture surfaces of H-SLM-WAAM and V-SLM-WAAM samples also appear very similar to those of WAAM samples because all the fractures are located in the WAAM area. The dimples with the size of 2–10 μm are deep, which also validate the ductile fracture of SLM-WAAM tensile specimens.

4. Conclusions

In this study, the hybrid SLM-WAAM fabrication process of Ti-6Al-4V is investigated. The SLM-WAAM samples with high density and good mechanical properties can be fabricated. The following conclu-

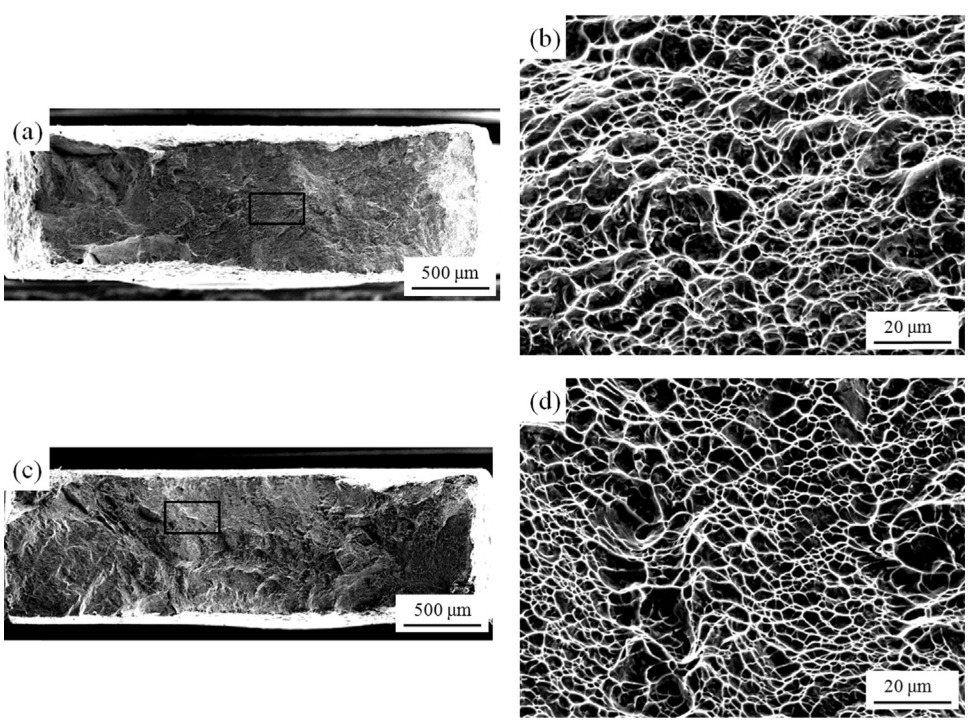

Fig. 12. SEM micrographs of tensile fracture surfaces. (a) Overall view of horizontal SLM-WAAM hybrid-processed tensile sample. (b) Enlarged view from the boxed region in "a." (c) Overall view of vertical SLM-WAAM hybrid-processed tensile sample. (d) Enlarged view from the boxed region in "c".

sions can be drawn:

1. Ti-6Al-4V sample formed by hybrid fabrication consists of three typical zones: SLM zone, WAAM zone and interface zone. There is a good metallurgical bonding in the interface zone of both H-SLM-WAAM and V-SLM-WAAM specimens.
2. Both horizontal section and vertical section SLM samples present fine and short β columnar grains consisting of primarily acicular martensite α´. The microstructure of interface zone is more complex, the fine β columnar grains from SLM zone grow coarse from 150 µm to 250 µm width in the about 3 mm-thick interface zone. In contrast, the coarse β columnar grains awash with α lamellae lead to epitaxial growth in the WAAM zone.
3. The mean yield strength, ultimate tensile strength and elongation of the horizontal SLM specimens are 1020 MPa, 1098 MPa and 10.2%, respectively. In comparison, the samples tested from the vertical direction exhibited slightly higher tensile strengths, with the yield strength of 1050 MPa, ultimate tensile strength of 1163 MPa and the elongation of 14.1%. By contrast, the WAAM samples have the lower strengths and ductility, the mean yield strength, ultimate tensile strength and elongation are 814 MPa, 927 MPa and 10.5%, respectively. Overall, the tensile properties of the WAAM-produced Ti-6Al-4V samples are slightly lower when compared to those of the SLM samples.
4. The mean yield strength, ultimate tensile strength and elongation of the H-SLM-WAAM specimens are 850 MPa, 905 MPa and 10.2%, respectively. In comparison, the V-SLM-WAAM samples exhibited slightly higher tensile strengths; the mean yield strength and ultimate tensile strength are 890 MPa and 995 MPa, respectively. The elongations of different directions are both more than 10%, satisfying the requirements of engineering applications. And all the fractures are located in the WAAM zone. The tensile properties of SLM-WAAM samples are comparable with or even better than the WAAM samples.

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